Fiscal Year:	FY 2022	**Task Last Updated:**	FY 08/25/2022
PI Name:	Wang, Clay Ph.D.		
Project Title:	Characterization of ISS Microorganisms That Assist in the Decomposition of Complex Organic Matter During Spaceflight		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Los Angeles	**State:**	CA
Zip Code:	90089-5305	**Congressional District:**	37
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG. App B: Flight and Ground Space Biology Research
Start Date:	09/01/2019	**End Date:**	08/31/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	August 2022 report: No change in personnel		
COI Name (Institution):	Singh, Nitin Ph.D. (NASA Jet Propulsion Laboratory) Venkateswaran, Kasthuri Ph.D. (NASA Jet Propulsion Laboratory)		
Grant/Contract No.:	80NSSC19K1501		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Title: Characterization of ISS microorganisms that assist in the decomposition of complex organic matter during spaceflight. Key objective of project: A broad array of experiments are performed continuously on International Space Station (ISS). These experiments generate waste matter that needs to be disposed of safely and cost-effectively, e.g., dead plant matter generated from the experiments carried out to check for the effect of microgravity on the development of roots and cells on plant seedlings. Filamentous fungi are used industrially as sources of enzymes to break down complex sugars from plant sources. The Wang group at the University of Southern California and Venkateswaran Group at Jet Propulsion Laboratory (JPL) have sent filamentous fungi on two different spaceflight projects (NASA Micro-10) and (CASIS-BRIC-NP). Our analysis of these flight samples compared to ground controls have shown differential expression particularly of genes related to carbohydrate metabolic processes. In addition, Microbial Tracking (MT-1) project isolated 202 microbial strains (138 bacteria and 64 fungi) and archived at JPL will be utilized to identify the microbial culture or consortium to degrade plant organic matter. This finding provides further justification that filamentous fungi and some unique bacterial strains can potentially assist in the decomposition of complex organic matter during spaceflight. The proposed investigation is a ground-based project to characterize microorganisms isolated from the ISS in the degradation of plant compost materials. We will utilize various simulated gravity systems available at JPL and NASA to understand the mechanism of how ISS microorganisms assist in the decomposition of complex organic matter during spaceflight. The objectives of the project and methods used are: 1) Identify the ISS microbial species isolated during MT-1 project which can degrade plant complex organic matter/ utilize complex sugar sources. 2) Genetically engineer up to five MT-1 microbial species (single or symbiotic combination of bacteria/fungi) down selected from objective 1 using CRISPR/Cas9 technology for creating mutants that are efficient in degrading complex organic matter. 3) Examine the proteomic profile of filamentous fungi and bacterial mutants created in objective 2 and wild type strains using simulated microgravity devices in comparison to ground control. 4) Examine the differential expression of carbohydrate degradation genes and their regulatory genes using simulated microgravity devices in comparison to ground control of mutants and wild type strains using whole genome sequencing and whole genome mRNA sequencing. Significance of proposed work: The proposed project addresses subtopic Pl-C ("Study changes in microbial populations from plants and plant growth media") experiments that demonstrate that symbiotic microbes can assist in the decomposition of complex organic matter during spaceflight. The project will not only provide insights into the mechanisms of microbial degradation of complex organic matter during spaceflight but aid in developing practical measures for mitigating risk to human health and ISS environment through waste disposal. The data generated will augment the NASA GeneLab database, which provides NASA with the statistical confidence to characterize microbes associated with closed habitation and crew health maintenance. The analysis proposed will help elucidate the impact of spaceflight stress on the production of novel metabolites and is directly responsive to the Space Biology Decadal Survey Recommendation, P1 (Use the ISS as a microbial observatory to determine foundational results that will be critical in the design of strategies to manipulate and control the microbiome of future exploration and contribute to the health of future space explorers). Furthermore, specific project objectives will enable us to answer Space Biology guiding questions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research will improve our understanding of plant degradation and improvement in the sustainable economy.
Task Progress:	Decomposition of Complex Organic Matter (DeCOM project) Progress report for 80NSSC19K1501 Clay C. C. Wang principal investigator Period covered to July 1, 2022 University of Southern California The DeCOM project is a ground-based investigation to characterize microorganisms isolated from the International Space Station (ISS) in the degradation of plant compost materials. We will utilize various simulated gravity systems available at the Jet Propulsion Laboratory (JPL) and NASA to understand the mechanism of how ISS microorganisms assist in the decomposition of complex organic matter during spaceflight. Current efforts will focus on the ground-based analysis for complex sugar/carbohydrate degradation by the fungal and bacterial isolates from the ISS. Objective 1: Identify the ISS microbial species isolated during the Microbial Tracking (MT-1) project which can degrade plant complex organic matter/utilize complex sugar sources a. Task 1. Selection of the ISS-isolated microbial species capable of growth on a wide range of carbon sources. b. Task 2. Chemical composition of complex sugars associated with plant wastes to resurrect "synthetic" non-degradable sugars. c. Task 3. Analysis of the ISS-isolated microbial species for their ability to degrade 15 different plant waste and generate sugar monomers. Objective 2: Genetically engineer up to five MT-1 microbial species (single or symbiotic combination of bacteria/fungi) down selected from Objective 1 using CRISPR/Cas9 technology for creating mutants that are efficient in degrading complex organic matter Objective 3: Examine the proteomic profile of filamentous fungi and bacterial mutants created in Objective 2 and wild-type strains using simulated microgravity devices in comparison to ground control Objective 4: Examine the differential expression of carbohydrate degradation genes and their regulatory genes using

	simulated microgravity devices, in comparison to ground control of mutants and wild-type strains using whole genome sequencing and RNA sequencing In the third grant period, we grew the ten down selected fungi for growth on lignin as the sole carbon source. We examined their metabolite profile using liquid chromatography-mass spectrometry. With the ethyl acetate extracts of the growth media, the fungal species can break down lignin and produce complex organic molecules. In Objective 2, we propose to engineer up to five MT-1 microbial species. In period 2, we created a genetic system for IMV00236, and the manuscript was published in 2021. In period 3, we developed genetic systems for two additional species, A. wentii and A. steynii, using CRISPR/Cas9 technology. The manuscript describing the CRISPR/Cas9-based genetic system for A. wentii has been submitted to ACS Chemical Biology, and a revision has been requested. We are addressing the issues the reviewers brought up, and the revised manuscript will be submitted in the next 60 days. A second manuscript describing A. steynii genetics is under preparation. Publications resulting from this grant period, including review publication and disclosure of techniques developed during this task period: 1. Bijlani, S., Parker, C., Singh, N.K., Sierra, M.A., Fox, J., Wang, C.C.C., Mason, C.E., Venkateswaran, K., Genomic Characterizatio of the Titan-like Cell Producing Naganishia tulchinskyi, the First Novel Eukaryote Isolated from the International Space Station. Journal of Fungi 2022; 8, 165. 2. Yuan, B., Keller, N.P., Oakley, B.R., Stajich, J.E., and Wang, C.C. C. Manipulation of the Global Regulator mcrA Upregulates Secondary Metabolite Production in Aspergillus wentii using CRISPR-Cas9 with in vitro-Assembled Ribonucleoproteins. In revision. 2022.
Bibliography Type:	Description: (Last Updated: 09/05/2022)
Articles in Peer-reviewed Journals	Bijlani S, Parker C, Singh NK, Sierra MA, Foox J, Wang CCC, Mason CE, Venkateswaran K. "Genomic characterization of the titan-like cell producing Naganishia tulchinskyi, the first novel eukaryote isolated from the International Space Station." J Fungi (Basel). 2022 Feb 8;8(2):165. https://doi.org/10.3390/jof8020165 ; PMID: 35205919; PMCID: PMC8875396 , Feb-2022